AI Critique

AI was most valuable as a coverage assistant, but it was unreliable as an authority for expected behavior. During FR-04 generation, it proposed oracles that were stronger than the specification, including assumptions about exact response behavior and Unicode handling. Human review marked such cases incomplete or invalid, corrected the observable invariants, and removed duplicate boundary maps and padding cases. The strongest AI contributions were security and state-transition ideas: role tampering in FR-04, cart-derived totals and cart clearing in FR-08, and authorization contexts plus persisted-state checks in FR-15.

The main failure pattern was confusing a plausible implementation with a documented contract. This also made coincidental PASS results dangerous. In FR-08, requests whose client total happened to equal the cart total passed even though the backend did not recalculate the value; a PASS therefore did not prove the intended computation. AI also needed explicit boundaries for security claims. Black-box injection requests cannot prove SEC-05's internal use of parameterized queries, and API read-back of HTML-like text cannot prove SEC-04 escaping at the later display boundary.

Real evidence still required human control. Newman output established observed behavior, but the student had to confirm genuine bugs, inspect screenshots, approve Issue publication, and distinguish product defects from harness defects and specification gaps. The lesson is that AI should expand the search space and structure traceability, while humans must own requirements, oracles, evidence, and final judgment. The collaboration worked best when every AI suggestion had a source anchor, one diagnostic purpose, a read-back oracle, and a stated limit on what the result could prove.